Reading Identity from Pixels: A Controlled Comparison of Eigenfaces and Deep Face Embeddings, with a Webcam Case Study on Illumination

Nic Groenewald
Independent project

2026

Abstract

Face recognition compares images by turning each into a vector and measuring distances. Two recipes produce the vectors: a classical linear subspace fitted to the pixels (Eigenfaces), and a deep network trained on millions of faces. This paper reports a controlled replication and measurement study of how much the second recipe buys, and where its claimed robustness to illumination comes from. On the 400-image Olivetti Faces dataset we compare a raw-pixel Eigenfaces map (4096-d) with a frozen VGGFace2-pretrained Inception-ResNet-v1 embedding (512-d, unit norm), using identical PCA settings for visualisation and computing every metric in full dimensionality. We add an open-set verification analysis under one protocol for both representations, and a webcam field study of one person under six lighting conditions.

The embedding wins on every identity metric: silhouette 0.106 to 0.497, 1-NN accuracy 94.5% to 100%, pair ROC area 0.881 to 0.9995, and open-set equal error rate 8.7% to 0.22% (at the protocol's 0.5% false-reject resolution). At a matched 1% false-reject rate, Map A accepts 48.6% of impostors against 0.044% for Map B. Two methodological findings matter as much. The raw-pixel 2-D plot preserves *more* of its space's pairwise geometry (rank correlation 0.807) than the embedding's (0.517), so the prettier picture belongs to the worse space. And "robust to lighting" conflates two effects: brightness changes cost nothing beyond repeat-scan noise, while light *direction* costs 3.83 times the frame-to-frame noise floor, with under-lighting worst. The work is a controlled replication, not a new method: Map A is fitted transductively on all 400 images and the deep model's training corpus is far larger than the test set, so the numbers measure a gap under favourable conditions.

1 Introduction

A face recognition system is a distance function wearing a decision rule. An image becomes a vector; two vectors are compared; a threshold turns the comparison into "same person" or "different person". Everything follows from the map: if one identity's vectors land close together and different identities land far apart, one threshold can work; if not, no threshold tuning will save it.

There are two canonical maps. Eigenfaces applies PCA to centred face images (Turk and Pentland, 1991): unsupervised, cheap, interpretable. The modern approach trains a deep network on a large corpus with a metric-learning objective—FaceNet's triplet loss (Schroff et al., 2015), VGGFace2's 3.31 million images (Cao et al., 2018), ArcFace's angular margin (Deng et al., 2019)—and uses it as a fixed feature extractor; NIST's FRVT evaluates such systems at scale (Grother et al., 2019). Does the learned embedding separate identity better, and can the difference be shown with a number

rather than a nice-looking picture? That question is usually answered either by a benchmark table that hides the mechanism or by two scatter plots that are an opinion. We wanted a comparison small enough to inspect completely and controlled enough that the difference cannot be attributed to plotting choices.

Framing. This is not a novel-method paper; it is a controlled replication and measurement study with a practitioner's case study attached. The comparison is not "learned versus raw" in any clean sense: it is a pretrained embedding trained on millions of faces versus a linear subspace fitted on the same 400 Olivetti images later used as probes. Map A is unsupervised but transductive—PCA sees every image, probes included—which if anything favours Map A. Map B invents detail upsampling 64×64 to 160×160, and its training corpus dwarfs the test set. The honest claim is conditional: under these conditions the pretrained embedding separates identity better, and the gap can be measured.

What was done. Three studies share one codebase and one measurement discipline. (i) *Two maps*: Olivetti's 400 greyscale images of 40 people (Samaria and Harter, 1994), processed as raw pixels (Map A) or frozen embeddings (Map B), with identical PCA settings for visualisation and metrics in full dimensionality. (ii) *One open-set verification protocol for both*: five enrol and five probe per identity, five seeded splits, reporting EER, TAR at fixed FAR, and FAR at matched 1% FRR. (iii) *A webcam illumination field study*: twelve scans of one person, ten frames each, under six lighting conditions—a case study of whether "robust to lighting" holds uniformly, and it does not.

Measurement discipline and findings. Two-dimensional PCA maps are for human eyes; a projection can look convincing while discarding the geometry a classifier uses, and our raw-pixel map preserves more of its true pairwise structure than the embedding's even though the embedding is better by every metric. We quantify the loss with a rank correlation between 2-D and full-dimensional pairwise distances and treat any statement not backed by a full-dimensional number as a hypothesis. The embedding wins every identity metric, and the margin grows as the task becomes more like a real decision: silhouette 0.106→0.497, 1-NN 94.5%→100%, pair ROC-AUC 0.881→0.9995, open-set EER 8.7%→0.22%. The webcam study shows the lighting invariance has a direction: brightness changes are indistinguishable from repeat-scan noise, while moving the light costs 3.83 times the noise floor, under-lighting worst. The cross-domain distance to Olivetti is large and domain-confounded; we report it as an upper bound only.

What is not claimed. No state-of-the-art performance, new embedding, or calibrated security system is claimed: the verification data are small and pre-aligned from a single session, and the webcam study has one subject and no stranger.

2 Background

Verification, identification, and Eigenfaces. *Identification* asks which enrolled identity a probe belongs to; *verification* asks whether it matches a claimed identity. Both need a representation ϕ and a distance d; verification compares $d(\phi(x), \phi(y))$ with a threshold τ. In the open-set setting the claimed identity may not be enrolled, so the system must say "unknown"—the setting of face unlock and of our verification experiment. Eigenfaces stacks each centred image as a vector in $\mathbb{R}^{4096}$, computes the leading eigenvectors of the covariance, and projects onto the first k (Turk and Pentland, 1991). The components can be reshaped into images, but the objective is variance, not between-identity separation, and on faces the largest variance is usually illumination. On tightly

cropped, roughly aligned data such as Olivetti it is nevertheless a strong baseline, and therefore the right comparator.

Deep face representations. DeepFace showed that a deep network trained on millions of faces could approach human verification performance (Taigman et al., 2014). FaceNet's triplet loss optimises the verification property directly: anchor–positive close, anchor–negative far by a margin (Schroff et al., 2015). VGGFace2 scaled the labelled corpus to 3.31 million images of 9,131 identities (Cao et al., 2018); ArcFace replaced the triplet objective with an additive angular margin (Deng et al., 2019); surveys trace the progression (Wang and Deng, 2021). Here the network is a fixed feature extractor: Inception-ResNet-v1 (Szegedy et al., 2017) pretrained on VGGFace2 via `facenet-pytorch` (facenet-pytorch, 2024), frozen, producing an L2-normalised 512-d vector. Freezing isolates what a pretrained representation contains without confounding the comparison with training that 400 images could not support. For unit vectors, cosine similarity is $1 - d^2_{\text{eucl}}/2$.

Detection and illumination. MTCNN is a cascade: a proposal network (P-Net) scans an image pyramid, a refinement network (R-Net) filters and regresses, and an output network (O-Net) refines and predicts five landmarks (Zhang et al., 2016). The `facenet-pytorch` implementation crops the O-Net box to a square and resizes to 160×160; it returns the landmarks but does not warp the patch, so alignment is detector-box alignment. Olivetti avoids the issue: its faces are pre-aligned, which makes it the best case for raw pixels and an easy case for the deep model. Illumination is the canonical nuisance: Extended Yale Face Database B held pose and expression roughly constant while sweeping light, and illumination-cone models showed that images under all lighting directions lie near a low-dimensional convex cone (Georghiades et al., 2001). Changing the *amount* of light scales pixel values roughly uniformly; changing the *direction* produces shading that moves across the face.

Metrics and visualisation. Verification is characterised by false accept rate (FAR), the fraction of impostor trials accepted, and false reject rate (FRR), the fraction of genuine trials rejected. Sweeping the threshold traces an ROC curve of TAR $= 1 -$ FRR against FAR; summary points are the equal error rate (EER), where FAR = FRR, and TAR at fixed FAR, which is how operational systems are compared because deployments fix the acceptable false-accept rate rather than the threshold (Grother et al., 2019). FERET established the vocabulary of defined gallery/probe splits and impostor trials (Phillips et al., 2000). We also use discriminability d' under an equal-variance Gaussian assumption,

$$d' = \frac{\mu_{\text{imp}} - \mu_{\text{gen}}}{\sqrt{(\sigma^2_{\text{gen}} + \sigma^2_{\text{imp}})/2}}. \tag{1}$$

PCA is a projection, and a projection can lie by omission; t-SNE is the usual cautionary example (van der Maaten and Hinton, 2008), but ordinary PCA scatters have a subtler problem: the two directions of largest variance need not be the two that separate identities. We quantify how much of the full-dimensional geometry a 2-D map preserves using the Spearman rank correlation between 2-D and full-dimensional pairwise distances, and never compute a metric on 2-D coordinates.

3 Related work

Eigenfaces (Turk and Pentland, 1991) is the reference appearance-based method; its linear descendants replaced PCA's unsupervised criterion with label-aware objectives, and its failure modes motivated decades of illumination normalisation. Our Map A is deliberately the plain 1991 recipe,

because the point is to measure what the raw representation contains. The deep lineage from DeepFace (Taigman et al., 2014) through FaceNet (Schroff et al., 2015), VGGFace2 (Cao et al., 2018), and ArcFace (Deng et al., 2019) is surveyed by Wang and Deng (2021); NIST's FRVT evaluates submitted systems on millions of images at fixed FAR (Grother et al., 2019). On measurement, the visualisation-versus-embedding distinction is well known (van der Maaten and Hinton, 2008), but the observation that the *worse* representation can have the more faithful 2-D plot is a useful cautionary result. Small, fully specified replications are valuable precisely because they are inspectable.

4 Data

Olivetti Faces. Olivetti (Samaria and Harter, 1994) provides 400 greyscale images of 40 people, ten each, at 64×64 pixels in $[0, 1]$. The images are tightly cropped and roughly eye-aligned: across one person's ten photographs the eyes sit in nearly the same place, and what changes is expression, head tilt, and lighting. This is the best case for raw pixels, because the geometric normalisation pixel methods are most sensitive to has already been done; it is also an easy case for a deep model. All Olivetti results should be read as an upper bound for the pixel representation. For the map comparison every image is used once—nothing is trained, so there is no split—and for verification the images are partitioned per identity into five enrolment and five probe images, identically for both maps. The dataset-wide pixel range is $[0, 1]$, and the preprocessing of the 400-image network batch matches the reference recipe exactly.

Webcam scans. Twelve scans of the author's face were captured on 6 September 2026 with a laptop webcam, ten consecutive frames each, each frame processed through the same MTCNN and embedding pipeline. The saved artefact per scan is a $(10, 512)$ float32 matrix of unit-norm embeddings; raw frames were not retained. Conditions: room light from the front and left, and phone flash from the front, left, right, and below, two repeat scans per condition. Room light was only available from two directions, so the design is unbalanced by necessity (twelve scans, not sixteen). Frames within a scan are consecutive webcam frames of a moving person, not independent photographs: mean embedding distance grows from 0.153 at lag 1 to 0.257 at lag 9, so the effective sample size per scan is below ten and scan means are the more meaningful unit. The first frame of a scan is also systematically different (mean distance 0.243 to the rest of the scan at frame 0 versus 0.167 at frame 5), consistent with webcam warm-up.

Model. The embedder is Inception-ResNet-v1 pretrained on VGGFace2 (Szegedy et al., 2017; Cao et al., 2018) via `facenet-pytorch` (facenet-pytorch, 2024). The checkpoint is `20180402-114759-vggface2.pt`, loaded in evaluation mode and frozen. It has 23,482,624 embedding parameters; the unused 8,631-class head adds 4,427,703. Outputs are L2-normalised. Nothing is trained or fine-tuned.

5 Method

5.1 System and preprocessing

Figure 1 shows the system. Olivetti images enter a raw-pixel branch (flattened 4096-d; PCA to 2-D for presentation) and a frozen-embedding branch (preprocess, embed, the same PCA settings for presentation); both feed a shared evaluation module that computes every metric in native dimensionality. The live path reuses the embedding branch with MTCNN on CPU and embedding on Apple MPS.

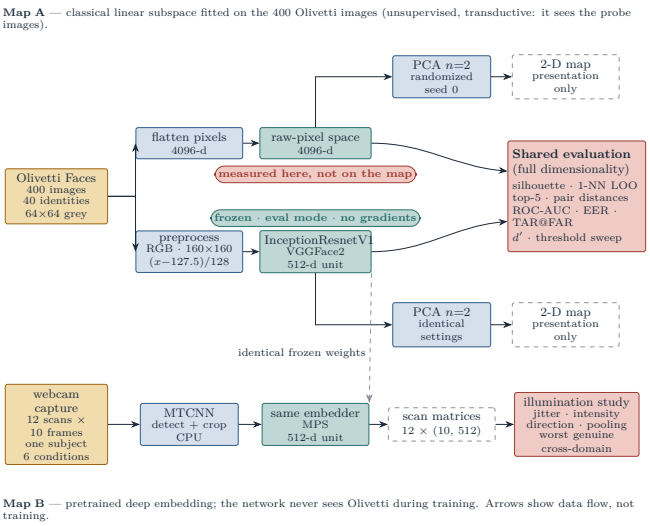


Figure 1: End-to-end system. Raw-pixel branch (Map A), frozen-embedding branch (Map B), and the live webcam path. The 2-D PCA maps are for inspection only; all metrics are computed on the full 4096-d and 512-d vectors before projection. Map A is fitted transductively on all 400 images; Map B's network never saw Olivetti during training.

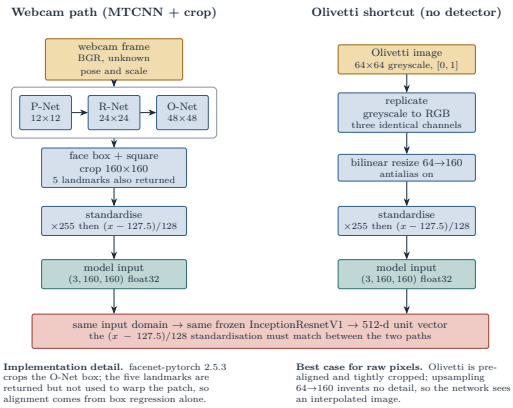


Figure 2: The two preprocessing paths. Webcam (left): MTCNN detects and crops, then the network's fixed standardisation is applied. Olivetti (right): no detector; greyscale is replicated to RGB, resized 64→160 with antialiasing, and standardised identically. The $(x - 127.5)/128$ step is part of the model definition.

Both paths must produce the same input domain, because the network's standardisation is part of the model (Figure 2). For Olivetti, each 64×64 greyscale image in $[0, 1]$ is replicated to three identical channels, bilinearly resized to 160×160 with antialiasing, multiplied by 255, and standardised by

$$x \mapsto \frac{x - 127.5}{128}. \tag{2}$$

The resize invents no detail, and the recipe reproduces the reference implementation exactly on the 400-image batch. For webcam frames, MTCNN detects a face box with a three-stage cascade, crops a square, resizes to 160×160, and applies the same standardisation. The library crops the O-Net box and does not warp on the five landmarks, so alignment is detector-box alignment; MTCNN runs on CPU because its operations are unsupported on MPS.

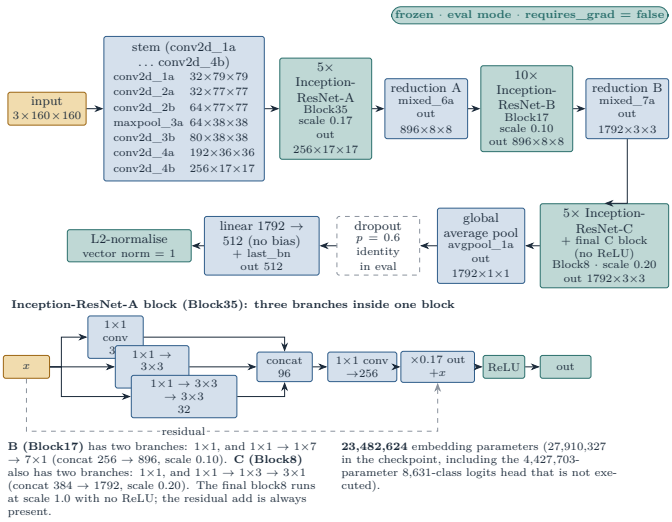


Figure 3: Inception-ResNet-v1 embedding network. Shapes are measured from `results/model_arch.json` (forward hooks on the VGGFace2 checkpoint, batch dimension dropped) and follow `facenet-pytorch` 2.5.3 naming. The model is frozen: evaluation mode, no gradients, 23,482,624 embedding parameters. The inset shows the Inception-ResNet-A block; B and C have two branches each, with kernel shapes and scales given in the note.

5.2 Frozen embedder and PCA maps

Figure 3 shows Inception-ResNet-v1 (Szegedy et al., 2017) with tensor shapes measured by forward hooks on the actual VGGFace2 checkpoint (Cao et al., 2018; facenet-pytorch, 2024). A seven-operation stem is followed by five Inception-ResNet-A blocks (Block35, scale 0.17) at $256 \times 17 \times 17$, reduction A (`mixed_6a`) to $896 \times 8 \times 8$, ten Inception-ResNet-B blocks (Block17, scale 0.10), reduction B (`mixed_7a`) to $1792 \times 3 \times 3$, five Inception-ResNet-C blocks (Block8, scale 0.20) plus a final C block without ReLU, global average pooling to 1792, a bias-free linear layer and batch normalisation to 512, and L2 normalisation. The model is in evaluation mode with gradients disabled, so embedding is deterministic up to hardware arithmetic, and embeddings are cached.

Both 2-D maps use PCA with two components, randomised SVD, seed 0, and sklearn's default centring (Pedregosa et al., 2011); the same settings, colours, and style are used for both. We deliberately do not apply `StandardScaler`: all 4096 pixel columns share one unit, so per-column scaling would inflate near-constant background pixels and change the variance ordering for reasons unrelated to identity; classic Eigenfaces centres but does not standardise. The same implementation produces the explained-variance analysis and the eigenface images; the 2-D coordinates are never a metric input. A matched-dimensionality control projects the raw pixels onto their full rank of 399 components (400 centred samples span at most rank 399), which preserves Euclidean distances exactly (maximum absolute difference 3.5×10^{-12}) and reproduces Map A's metrics to numerical precision. The raw representation is intrinsically 399-dimensional here, so the silhouette and EER gaps are not dimension artefacts.

5.3 Metrics and protocols

Distances are Euclidean; for unit vectors, cosine similarity is $1 - d^2/2$. The silhouette score is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \qquad S = \frac{1}{N} \sum_{i=1}^{N} s(i), \tag{3}$$

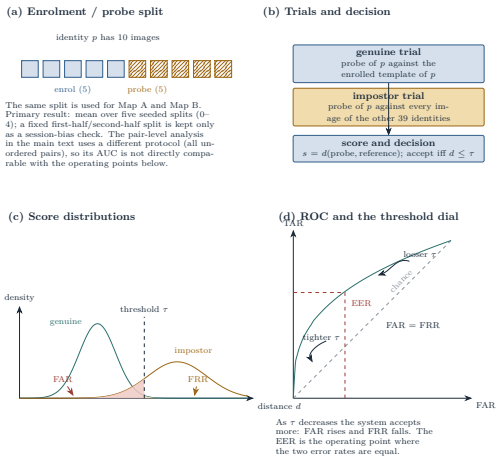


Figure 4: Evaluation protocol. (a) Per identity, five images enrol and five probe; the split is identical for both maps. (b) Genuine and impostor trial definitions. (c) Score distributions with threshold τ; FAR is impostor mass at or below τ, FRR is genuine mass above it. (d) ROC curve and the security–convenience dial: lowering τ admits more genuine probes and more impostors. Curves in (c) and (d) are schematics, not measured data.

with $a(i)$ the mean distance to the same identity and $b(i)$ the mean distance to the nearest other identity, on full-dimensional vectors. *1-NN leave-one-out* finds each image's nearest neighbour among the other 399, excluding self; top-5 asks whether the true identity is among the five nearest. *Pair-level* analysis scores all $\binom{400}{2} = 79{,}800$ unordered pairs: 1,800 genuine (40 identities $\times\binom{10}{2}$) and 78,000 impostor. FAR and FRR at threshold τ are

$$\mathrm{FAR}(\tau) = \frac{\#\{d_{\mathrm{imp}} \le \tau\}}{N_{\mathrm{imp}}}, \qquad \mathrm{FRR}(\tau) = \frac{\#\{d_{\mathrm{gen}} > \tau\}}{N_{\mathrm{gen}}}, \tag{4}$$

from which we report ROC-AUC, EER and its threshold, TAR at FAR $= 10^{-3}$ and 10^{-2}, and d' from Equation 1. *2-D fidelity* is the Spearman rank correlation between 2-D and full-dimensional pairwise distances over 20,000 seeded pairs. The *confusion analysis* compares 1-NN predictions across maps; with zero Map B errors the "both wrong" fraction has an empty denominator and is `null`, not zero.

The map comparison uses every image once. Verification uses the open-set protocol in Figure 4: for each of 40 identities, five images enrol and five are genuine probes; a probe's score is its *minimum* distance to the five enrolment images, and the decision is accept if the score is at most the threshold. Impostor trials score each probe against the 390 images of the other 39 identities. The primary result averages five random 5/5 splits (seeds 0–4): 200 genuine and 15,600 impostor trials per split, identical for both representations. We report per-split EER, TAR at FAR $= 10^{-3}$ and 10^{-2}, AUC, and FAR at matched 1% FRR, plus a fixed first-half/second-half split (session-bias check) and a mean-template score rule (sensitivity check).

5.4 Live protocol and measurement rule

The twelve scans are analysed at frame and scan level (Figure 5). *Within-scan jitter* is the mean of all 45 frame-pair distances inside a scan: the frame-to-frame noise floor. *Same-condition between-scan* distance compares the repeat scans of each condition. The *intensity effect* compares room-front with flash-front frames; the *direction effect* is the mean over the six flash direction pairs, and the headline ratio divides it by the noise floor. *Pooling* compares frame-level distances with distances between

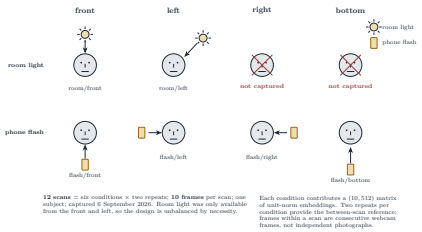


Figure 5: Webcam capture conditions. Room light and phone flash were used from up to four directions; room light was only available from front and left, so the design is unbalanced by necessity. Each captured condition has two repeat scans of ten consecutive frames; early frames are affected by webcam warm-up.

re-normalised scan means. The *worst genuine* pair is the largest same-person frame-pair distance, with its cosine equivalent and the same quantity after scan averaging. The *cross-domain* check measures distances from the 400 Olivetti embeddings to the live scans; it is domain-confounded and reported as an upper bound only.

Measure in full dimensionality, plot in 2-D: no metric is computed on PCA coordinates, and the rank correlation quantifies what the plot loses. *One pipeline*: load, embed, project, plot, score, with only the representation swapped. All experiments use seed 0 unless a protocol averages seeds 0–4; every reported number is read from machine-written JSON. Hardware and versions are in Section 9.

6 Results

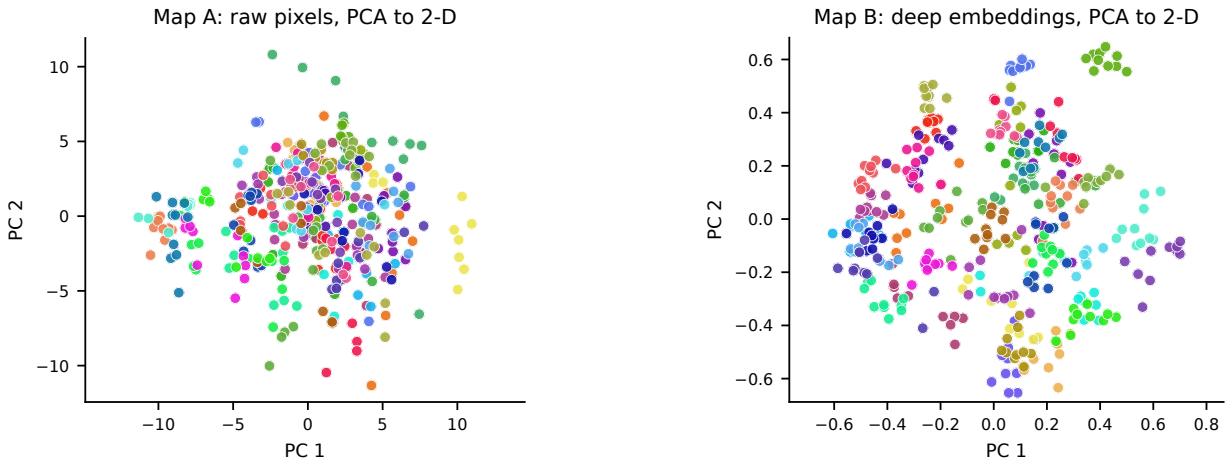
6.1 The two maps

Figure 6 shows the two 2-D PCA maps with identical plotting settings. Map A's leading components are brightness gradients, with 23.8% of pixel variance in component 0 and 2, 4, 19, and 66 components for 25%, 50%, 75%, and 90% (95% not reached in the first 100); Map B needs 3, 7, 15, 24, and 31, and its first component carries 10.9%. Map A's 2-D axes are illumination: PC1 is a brightness axis ($r = 0.986$ with mean image brightness) and PC2 a lateral gradient ($r = 0.944$); the lobes are brightness groups, not identities.

The crucial check is the rank correlation between 2-D and full-dimensional pairwise distances: Map A's plot preserves more of its space's geometry ($\rho = 0.807$) than Map B's plot preserves of the embedding space ($\rho = 0.517$). The embedding is the better representation; its 2-D picture is the less faithful one. Figure 7 shows the eigenfaces and variance curves. The projections are stable across seeds 0–2, while full-dimensional metrics are deterministic.

6.2 Quantitative comparison

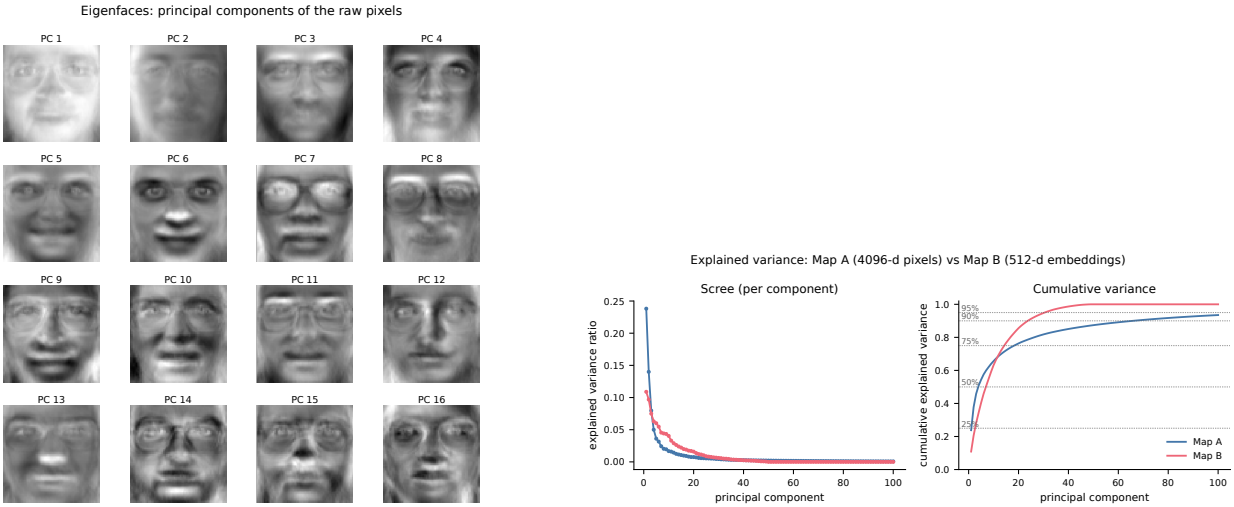
Table 1 gives the full-dimensional comparison; every row favours the embedding. Silhouette more than quadruples, 1-NN accuracy reaches 100% from 94.5%, pair ROC-AUC rises from 0.881 to 0.9995, pair EER falls from 20.5% to 1.11%, and d' roughly triples. Map A's genuine distances (8.38 ± 2.56) overlap heavily with its impostors (12.41 ± 2.45), while Map B's are nearly disjoint (0.53 ± 0.19 versus 1.33 ± 0.13; Figure 8). The matched-dimensionality control (399 components, 100% variance retained) reproduces Map A's metrics exactly (silhouette 0.10557, AUC 0.8807, EER 0.2050; maximum distance difference 3.5×10^{-12}), so the gaps are not dimension artefacts.



(a) Map A: raw pixels.

(b) Map B: frozen embeddings.

Figure 6: The two 2-D maps with identical colour, axis, and PCA conventions. Presentation only: every metric is computed on the full 4096-d and 512-d vectors. The raw-pixel map preserves more pairwise geometry (rank correlation 0.807) than the embedding map (0.517), even though the embedding is the better representation.



(a) First 16 Eigenfaces.

(b) Explained and cumulative variance.

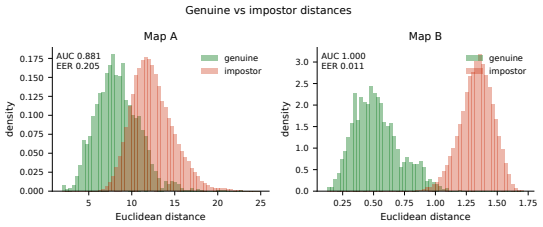
Figure 7: Map A's leading components are illumination and low-frequency shape patterns, not identity features; component 0 alone carries 23.8% of the raw-pixel variance. Map B's variance is spread over more components (31 for 95%) than Map A's concentrated variance.

6.3 Why the metrics disagree

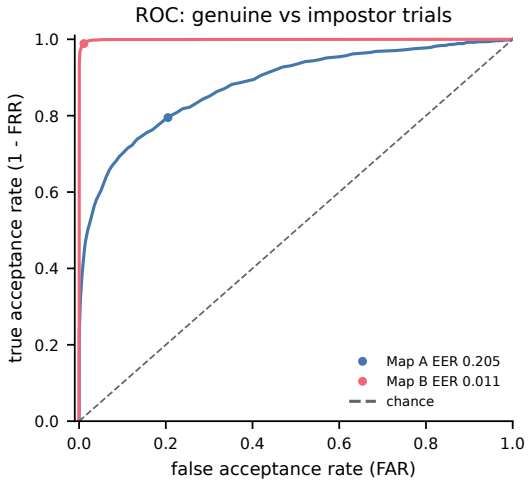
Raw pixels look respectable on identification (94.5% 1-NN) but poor on silhouette (0.106 versus 0.497) and worse on open-set verification (EER 8.7% versus 0.22%). Identification asks whether the single nearest gallery image is correct, and with ten images per identity and pre-aligned data it often is. Verification fixes one threshold and requires the distributions to be separated, not merely ordered; Map A's overlap makes that impossible, Map B's near-disjoint distributions make it easy. A good nearest-neighbour result is not evidence that a face-unlock threshold exists. The maps

Table 1: Main comparison in full dimensionality. Pair-level metrics use all 1,800 genuine and 78,000 impostor pairs from the 400 Olivetti images. The 399-component dimensionality control reproduces every Map A row to numerical precision.

Metric	Map A: raw pixels (4096-d)	Map B: embedding (512-d)
Silhouette (40 identities)	0.1056	**0.4970**
1-NN leave-one-out accuracy	0.9450	**1.0000**
1-NN leave-one-out top-5	0.9825	**1.0000**
Pair ROC-AUC	0.8807	**0.99954**
Pair EER	0.2050	**0.01111**
TAR at FAR $= 10^{-3}$	0.2639	**0.9550**
TAR at FAR $= 10^{-2}$	0.4233	**0.9883**
d'	1.612	**4.988**
Genuine distance (mean $\pm$ sd)	8.378 ± 2.556	0.5284 ± 0.1860
Impostor distance (mean $\pm$ sd)	12.414 ± 2.450	1.3336 ± 0.1324
2-D rank correlation	**0.8070**	0.5174



(a) Genuine and impostor distances.



(b) Pair-level ROC with EER.

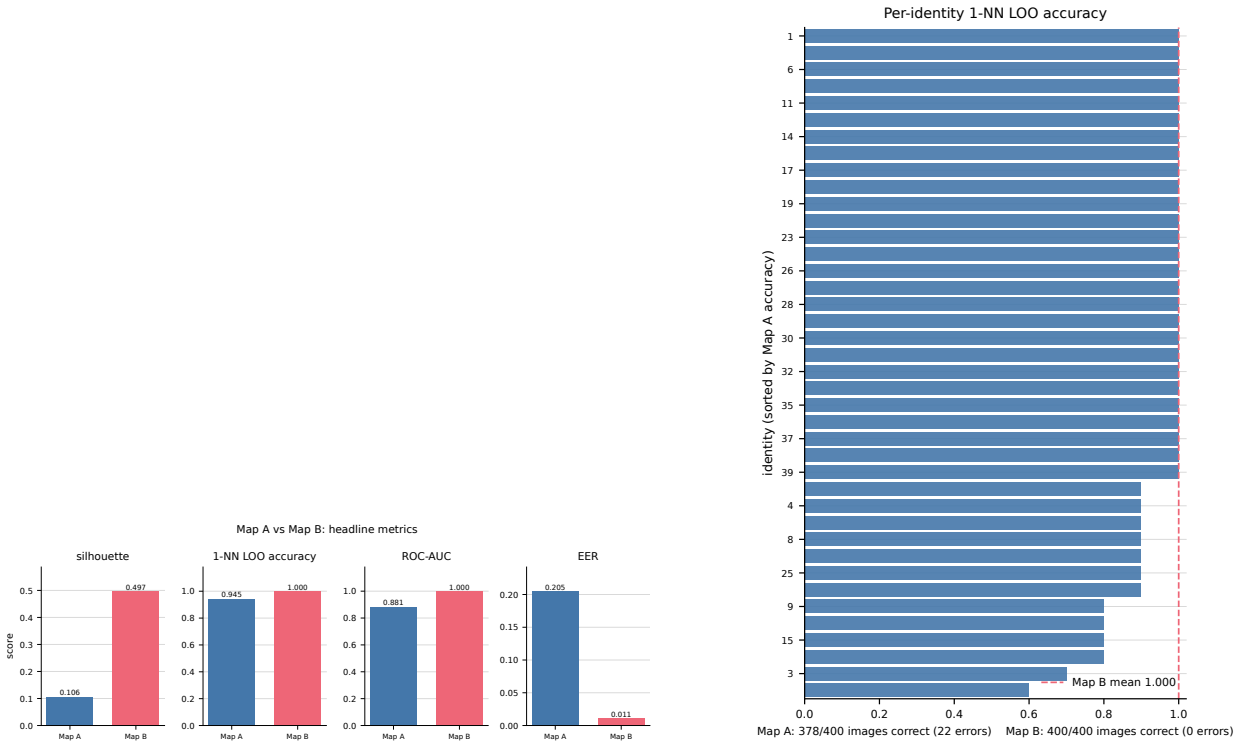
Figure 8: Pair-level verification. Map A's distributions overlap substantially; Map B's are nearly disjoint. Curves use the same 1,800 genuine and 78,000 impostor distances, not the 2-D maps.

agree on 378 of 400 predictions (94.5%), Map B makes no errors, and there is no "both wrong" set (reported as `null`, not zero). Figure 9 summarises the metric gaps and per-identity accuracy.

6.4 Open-set verification and the threshold dial

Table 2 reports the FaceID-style protocol. Map A's EER is 8.71% (sd 1.25 points) and Map B's is 0.22% (sd 0.24 points). Map B's figure is at or below the protocol's 0.5% FRR resolution (200 genuine trials per split), so read it as "at the measurement floor"; the roughly 39-fold EER ratio inherits that floor. The matched-FRR comparison is not floor-limited (15,600 impostor trials per split, FAR resolution 0.0064%): at 1% false-reject rate, Map A accepts 48.59% of impostors and Map B 0.044%, roughly 1,115 times fewer for Map B. At a 1% false-accept rate, Map B accepts 99.9% of genuine probes versus 74.6% for Map A.

Figure 10 shows the curves on separate axes: the EER thresholds are 8.505 and 0.8396, roughly ten times apart, and a shared axis would invite a false comparison of threshold values. Map A has



(a) Metric summary. **(b)** Per-identity 1-NN accuracy.

Figure 9: Metric summary and per-identity accuracy. The embedding wins every metric; Map B is perfect on every identity, while Map A's errors concentrate in a few identities. Cross-map prediction agreement is 94.5%.

Table 2: Open-set verification: five random 5/5 splits (seeds 0–4), 200 genuine and 15,600 impostor trials per split. Scores are minimum distances to the five enrolment images; accept if score $\leq \tau$. Mean (sd) across splits. Map B's EER is at the protocol's 0.5% FRR resolution.

Operating point	Map A	Map B
EER	0.0871 (0.0125)	0.00224 (0.00244)
EER threshold	8.505	0.8396
TAR at FAR $= 10^{-3}$	0.593	0.994
TAR at FAR $= 10^{-2}$	0.746	0.999
FAR at matched FRR $= 1\%$	0.4859 (0.1189)	0.000436 (0.000378)
AUC	0.96999	0.99997

no threshold with both error rates small; Map B has a broad operating region. Robustness checks do not change the ranking: the fixed first-half/second-half split gives EERs 9.00% and 0.256%, the mean-template rule 11.85% and 0.22%. Pair-level EERs (20.5% and 1.11%) are a different protocol and agree on ranking, not absolute value; the open-set protocol supplies the headline numbers.

6.5 Webcam illumination study

Table 3 summarises the study. The frame-to-frame noise floor is 0.186 (per-scan means 0.114–0.334), and same-condition repeats differ by 0.227–0.371. Changing the *amount* of light moves the mean distance to 0.258 (range 0.168–0.384), inside the repeat-scan range: brightness is effectively free. Changing the *direction* moves it to 0.714, 3.83 times the noise floor. Front-to-side pairs are large

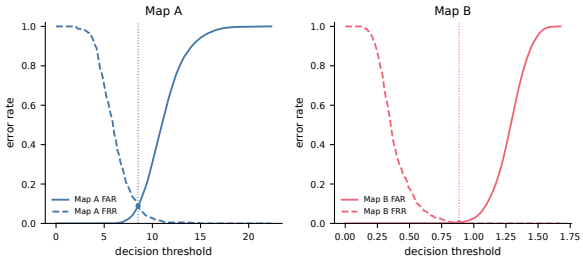


Figure 10: FAR (solid) and FRR (dashed) against threshold, with EER marked. Separate axes are used because the representations produce distances on different scales (EER thresholds 8.505 versus 0.8396). Map A has no threshold with both error rates small; Map B has a wide operating region.

Table 3: Webcam illumination study summary (Euclidean distances between unit-norm embeddings; cosine similarity $= 1 - d^2/2$). The noise floor is the mean within-scan frame-pair distance over twelve scans. The worst genuine pair is flash/right repeat 2 versus flash/bottom repeat 2. The cross-domain row is domain-confounded and is an upper bound only.

Quantity	Value
Within-scan jitter (noise floor)	0.186
Same-condition between-scan means	0.227–0.371
Intensity effect (room/front vs flash/front)	0.258
Direction effect (six flash direction pairs, mean)	0.714
Direction / noise ratio	3.83
Pooling reduction (scan means vs frames)	4.82%
Worst genuine frame pair	0.967
Worst genuine after scan averaging	0.915
Cross-domain minimum (Olivetti to live scan)	0.887

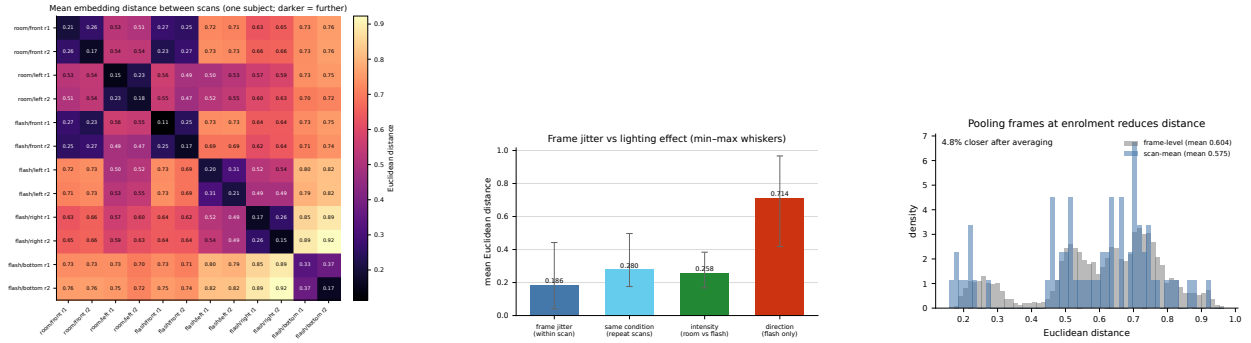
(0.636–0.733) while left–right is smaller (0.511), so two half-shadowed faces resemble each other more than either resembles a front-lit face—a "half-shadowed" region. Side-to-bottom is worst (0.808–0.888), consistent with under-lighting being rare in photographic training data. Pooling frames reduces mean distances by 4.82% (0.604 to 0.575), a modest gain that does not change condition ordering (Figure 11).

The worst same-person pair is flash/right repeat 2 against flash/bottom repeat 2: 0.967 frame-level (cosine similarity 0.532, cosine distance 0.468), 0.915 after scan averaging (cosine similarity 0.581, cosine distance 0.419). A frame-level threshold below 0.967 would reject the enrolled user in at least one observed condition; a scan-mean threshold below 0.915 would do so. That is the convenience side of the dial, measured on one person. Frames are not independent (mean within-scan distance grows from 0.153 at lag 1 to 0.257 at lag 9), so the effective sample size per scan is below ten; frame 0 is also a warm-up outlier (0.243 versus 0.167 at frame 5).

The smallest distance from any Olivetti embedding to any live scan is 0.887 (frame-level 0.880; mean of per-Olivetti minima 1.288). It is domain-confounded—different capture device, crop, resolution, and people—so it is an upper bound only, not a calibrated impostor distribution (Section 7).

7 Discussion

What the comparison shows. The embedding dominates every identity metric, but the interpretation is narrower than "deep beats classical". Map A is a plain linear subspace fitted on the



(a) 12×12 scan distances.

(b) Noise floor, intensity, direction.

(c) Frame-level vs scan means.

Figure 11: Live illumination study. Repeat scans of the same condition are close and condition blocks are separated by roughly the direction effect; brightness changes are indistinguishable from repeat-scan noise, while direction changes are about four times the frame-to-frame noise floor. Pooling ten frames into a scan mean reduces distances modestly and does not remove the direction effect.

same 400 images it is tested on, with no labels, illumination normalisation, or per-pixel scaling; it is shown at its best geometric conditions. Map B is a 23.5-million-parameter network trained on 3.31 million images and then frozen. The gap therefore combines representation, training corpus, and architecture, and this study cannot separate them; what it establishes is the gap's size under a controlled protocol, and that the transductive fit favours Map A, so the measured advantage of Map B is conservative.

The metrics disagree, and that is the point. Raw pixels look respectable on identification (94.5% 1-NN) but poor on silhouette and substantially worse on open-set verification. Identification asks whether the single nearest gallery image is correct, and with ten images per identity and pre-aligned data the nearest genuine image is often close enough. Verification fixes one threshold and requires the genuine and impostor distributions to be separated, not merely ordered. Map A's overlap makes that impossible; Map B's near-disjoint distributions make it easy. A good nearest-neighbour result is not evidence that a face-unlock threshold exists.

The prettier map is the worse space. Map A's 2-D plot is a more faithful projection of its own space ($\rho = 0.807$) than Map B's plot is of the embedding space ($\rho = 0.517$), and Map A's 2-D axes are literally brightness and a lateral lighting gradient. A viewer comparing the two scatters is not comparing recognition ability: the plots preserve unequal fractions of the geometry they summarise. Raw-pixel variance is dominated by a few illumination directions, so two components capture much of its structure even though that structure is not identity; the embedding's leading variance directions are less identity-discriminative, so compressing it discards more of what makes it useful.

Illumination: intensity is free, direction is expensive. Changing the amount of light moves the mean distance to 0.258, inside the 0.227–0.371 repeat-scan range: the model is effectively invariant to brightness. Changing the direction moves it to 0.714, or 3.83 times the noise floor, with side-to-bottom pairs worst. Left–right (0.511) is smaller than front–side pairs, so two half-shadowed faces resemble each other more than either resembles a front-lit face; the embedding appears to have a "half-shadowed" region rather than a simple left/right axis. For enrolment, a single front-lit scan

leaves the system exposed to side-lit probes; adding a side-lit scan would cover both half-shadowed regions—a hypothesis from one subject, not a validated policy.

The security–convenience dial. At matched 1% FRR, Map A accepts 48.6% of impostors and Map B 0.044%; at 1% FAR, Map B accepts 99.9% of genuine probes against 74.6% for Map A. Map B has a broad usable threshold range; Map A has none. Thresholds do not transfer: EER thresholds are 8.51 for pixels and 0.84 for embeddings, and webcam distances live on a third scale. The cross-domain check makes the point: the smallest Olivetti-to-webcam distance is 0.887 while the worst same-person frame pair is 0.967. Taken literally, a threshold accepting every genuine webcam frame would also accept the closest Olivetti face—but that number is domain-confounded and is an upper bound, not a calibrated impostor score. Domain shift is large enough to make a threshold fitted on one dataset meaningless on another; any deployment must calibrate on in-domain impostors, which this study cannot provide because it has no stranger capture.

8 Limitations

One subject, no stranger. The webcam study has one subject and no stranger was captured. Every live distance is same-person: it measures convenience, not security. A usable threshold needs an impostor distribution, which this study does not have; the cross-domain Olivetti-to-webcam distance is the closest thing to an impostor number and is explicitly domain-confounded.

Olivetti is the best case for raw pixels. The dataset is tightly cropped and roughly eye-aligned, removing the geometric variability that most damages pixel-space methods. Map A's results are an upper bound for raw pixels, and the same property makes the deep model's task easy, so both maps' absolute accuracy is optimistic.

Small data, weak statistics. 400 images, 40 identities, ten images per identity, all from a single session. The 1,800 genuine pairs are not independent (each image appears in nine pairs), and the five splits reuse the same images, so split variance is not capture variance. Map B's 0.22% EER is at the protocol's 0.5% FRR resolution and should not be read as three significant figures.

Transductive fitting and frozen model. Map A's PCA is fit on all 400 images, including the probes, which favours Map A; the measured advantage of Map B is therefore conservative. The embedder was not trained or fine-tuned on either dataset, which isolates the representation but says nothing about a model trained on Olivetti or webcam data, and its training corpus is far larger than the evaluation data.

Webcam frames are not independent. The ten frames per scan are consecutive webcam frames of a moving person: within-scan distance grows from 0.153 at lag 1 to 0.257 at lag 9, and the first frame is a warm-up outlier. Frame-pair counts overstate the effective sample size, which is why scan means are the more meaningful unit. Raw webcam frames were not retained, so alternate crops, detectors, or preprocessing cannot be re-analysed from the saved data.

Detection and thresholds. The MTCNN implementation crops the O-Net box without a landmark similarity transform, so alignment is detector-box alignment; a landmark-warped pipeline might reduce the direction effect, which this study does not test. Olivetti verification thresholds (EER thresholds 8.51 and 0.84) and the webcam distance scale are not comparable, and the cross-domain minimum (0.887) is domain-confounded. No threshold in this paper should be deployed.

No demographic, sensor, or session diversity. One dataset with limited demographic variety, one laptop webcam, one model/backbone, no repeated sessions across days. The optional synthetic-lighting ablation was not run, so the illumination analysis is observational and confined to the captured conditions.

9 Reproducibility

All experiments ran on an Apple M5 Pro (18 cores, 48 GB unified memory) under macOS 27.2 (arm64), Python 3.12.14, NumPy 2.5.3, scikit-learn 1.9.1, PyTorch 2.14.0, `facenet-pytorch` 2.5.3, and Matplotlib 3.11.2. The embedder runs on Apple MPS; MTCNN runs on CPU because its operations are unsupported by MPS. No training runs or cloud GPUs were used. Every experiment uses seed 0 unless a protocol averages seeds 0–4, and every sklearn estimator receives `random_state=0`. The CNN is deterministic in evaluation mode, and its Olivetti embeddings are cached. Results are written to JSON by the analysis scripts; the paper's numbers are read from those files, and each results file records the git commit, Python and package versions, device, and seed.

The pipeline is:

```
export MPLCONFIGDIR="$PWD/.cache/mpl"
PY=/Users/nic/.venvs/nn-crew/bin/python
$PY analysis/run_olivetti.py   # maps, metrics, model arch
$PY analysis/run_live.py       # webcam study
$PY analysis/run_threshold.py  # open-set verification
$PY analysis/model_arch.py     # layer shapes and param counts
$PY -m pytest -q               # fast tests, no network/webcam
cd paper && pdflatex -interaction=nonstopmode main.tex
```

The checkpoint is `20180402-114759-vggface2.pt`, cached locally (111,898,327 bytes; SHA-256 begins `281cebca8662831a987a874bdcb36e73`). Its measured layer shapes and parameter counts are in `results/model_arch.json`; Figure 3 is generated from those measurements. Olivetti Faces is public and ships with scikit-learn. The webcam embeddings, results JSON files, and figure-generation code are in the project repository; raw webcam frames were not retained. The analysis code is a library (`fem/`) plus experiment runners (`analysis/`), so the comparison re-runs without duplicating preprocessing or metrics. Small numerical differences between BLAS implementations are possible.

10 Future work

Yale B. Extended Yale Face Database B (Georghiades et al., 2001) sweeps lighting around the room for roughly 38 people at fixed pose. Re-running the two-map comparison there would test the prediction this paper can only sketch—raw pixels should cluster by lamp position, the embedding should remain identity-driven—with enough subjects to support a claim about people rather than one face.

LFW and alignment. Olivetti is the best case for raw pixels, so the comparison should be repeated on unconstrained data such as LFW (Huang et al., 2007). An intermediate step is an MTCNN-augmented Olivetti ablation: Map A on detector-cropped rather than pre-aligned images, and Map B on both crops, separating alignment from representation.

Training and representations. The project plan reserved training for a final stage; it remains out of scope. Training a small embedding with triplet loss on Olivetti would show how much of the

gap is available from the data at hand without a large pretrained corpus, and would require held-out identities. Repeating it with ArcFace (Deng et al., 2019) would test whether the conclusions are properties of deep embeddings generally or of this checkpoint.

More subjects, strangers, sessions. Several subjects, stranger images for impostor calibration, repeated sessions across days, and different sensors would turn the live study into a calibration experiment and let the security–convenience dial be set with data rather than described.

Synthetic relighting. Applying smooth illumination gradients to Olivetti images and re-embedding them would isolate direction from capture artefacts and connect the webcam observations to the illumination-cone literature.

References

Q. Cao, L. Shen, W. Xie, O. M. Parkhi, and A. Zisserman. VGGFace2: A dataset for recognising faces across pose and age. In *13th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2018)*, pages 67–74, 2018. doi:10.1109/FG.2018.00020; arXiv:1710.08092.

J. Deng, J. Guo, N. Xue, and S. Zafeiriou. ArcFace: Additive angular margin loss for deep face recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4685–4694, 2019. doi:10.1109/CVPR.2019.00482; arXiv:1801.07698.

`facenet-pytorch`: PyTorch face recognition models (MTCNN, InceptionResnetV1, VGGFace2/CASIA-WebFace weights), version 2.5.3. Software, `https://github.com/timesler/facenet-pytorch`, 2024.

A. S. Georghiades, P. N. Belhumeur, and D. J. Kriegman. From few to many: Illumination cone models for face recognition under variable lighting and pose. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 23(6):643–660, 2001. doi:10.1109/34.927464.

P. Grother, M. Ngan, and K. Hanaoka. Face Recognition Vendor Test (FRVT) Part 2: Identification. NIST Interagency/Internal Report (NISTIR) 8271, National Institute of Standards and Technology, Gaithersburg, MD, September 2019. doi:10.6028/NIST.IR.8271.

G. B. Huang, M. Ramesh, T. Berg, and E. Learned-Miller. Labeled Faces in the Wild: A database for studying face recognition in unconstrained environments. Technical Report 07-49, University of Massachusetts, Amherst, October 2007.

F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

P. J. Phillips, H. Moon, S. A. Rizvi, and P. J. Rauss. The FERET evaluation methodology for face-recognition algorithms. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(10):1090–1104, 2000. doi:10.1109/34.879790.

F. S. Samaria and A. C. Harter. Parameterisation of a stochastic model for human face identification. In *Proceedings of the Second IEEE Workshop on Applications of Computer Vision (WACV)*, pages 138–142, 1994. doi:10.1109/ACV.1994.341300.

F. Schroff, D. Kalenichenko, and J. Philbin. FaceNet: A unified embedding for face recognition and clustering. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 815–823, 2015. doi:10.1109/CVPR.2015.7298682; arXiv:1503.03832.

C. Szegedy, S. Ioffe, V. Vanhoucke, and A. Alemi. Inception-v4, Inception-ResNet and the impact of residual connections on learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 31, 2017. doi:10.1609/aaai.v31i1.11231; arXiv:1602.07261.

Y. Taigman, M. Yang, M. Ranzato, and L. Wolf. DeepFace: Closing the gap to human-level performance in face verification. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1701–1708, 2014. doi:10.1109/CVPR.2014.220.

M. Turk and A. Pentland. Eigenfaces for recognition. *Journal of Cognitive Neuroscience*, 3(1):71–86, 1991. doi:10.1162/jocn.1991.3.1.71.

L. van der Maaten and G. Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, 2008.

M. Wang and W. Deng. Deep face recognition: A survey. *Neurocomputing*, 429:215–244, 2021. doi:10.1016/j.neucom.2020.10.081; arXiv:1804.06655.

K. Zhang, Z. Zhang, Z. Li, and Y. Qiao. Joint face detection and alignment using multitask cascaded convolutional networks. *IEEE Signal Processing Letters*, 23(10):1499–1503, 2016. doi:10.1109/LSP.2016.2603342; arXiv:1604.02878.